Israel Souza Ferreira

Machine Learning Engineer | Data Scientist
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Professional Summary

Computer Science graduate building ML systems and evaluating models. Work spans computer vision, data ingestion, and graph optimization, with measured performance and documented limitations.

Education

Bachelor of Science in Computer Science
Federal University of Ceará (UFC) --- Quixadá Campus, Brazil
2021 -- August 2026; completed August 19, 2026.
Thesis in graph optimization with GA, ACO, and integer linear programming; defended February 2025.

Technical Skills

Programming and data: Python, SQL, C++, Pandas, NumPy, PostgreSQL/pgvector.
ML and evaluation: scikit-learn, PyTorch, OpenCV, ONNX Runtime, cross-validation, calibration.
Engineering: FastAPI, Docker, GitHub Actions, pytest, SQLAlchemy, Alembic.

Professional Experience

Research and Development (R&D) Researcher
Tieta.ai --- Remote, Brazil | July 2026 -- Present

- Develop and maintain research software with validation, automated tests, and versioned documentation.

Full Stack Developer and Machine Learning Engineer
Independent consulting for an educational institution --- Remote, Brazil | March 2026 -- Present
Face Clock Evoluir: v1.3.0-rc.3 is a supervised institutional pilot candidate deployed to a test environment.

- Built 1:N facial recognition, FastAPI APIs, and PostgreSQL/pgvector persistence; recalibration reduced unknown people accepted from 45.4% to 0.2% in a controlled test, with institutional calibration still pending.
- Used benchmarks to reduce frames and retain exact search (0.076 ms/1,000 vectors; tested HNSW recall@1 0.915); fixed extraction failures under concurrency.
- Implemented a seven-step institutional setup, audit trail, and HR workflows; prepared three release candidates and deployed to a test environment.

Machine Learning Researcher
Tieta.ai --- Remote, Brazil | April 2025 -- June 2026

- As a part-time CNPq RHAE fellow, developed and evaluated applied ML solutions with reproducible experiments.

UFC teaching assistant: Algorithm Design and Analysis (March--November 2023) and Calculus (April--July 2022).

Selected Projects

Triple Roman Domination in Graphs --- Undergraduate Thesis | 2024--2025

- Implemented GA and ACO in C++ and an ILP model in Python/Pyomo; a counterexample exposed an invalid published formulation. Evaluated 362 graphs, one run per instance.

Code: https://github.com/isrreal/Triple-Roman-Domination-in-graphs

Brazilian Emergency Aid --- Data Engineering | In development

- Built chunked Pandas and PostgreSQL binary COPY ingestion for a 31.6 GB public source. On subsets up to 5 million rows, post-cleanup RSS grew from 182 to 750 MiB; retained IDs correlated at r 0.991, without proving causality.

Code: https://github.com/isrreal/AuxilioEmergencialQueries

Languages

Portuguese: native. English: good communication; fluent technical reading of research papers and documentation.